

ADVANCED PROGRAMMING IN WEB TECHNOLOGY [A]

Name: MD. Nazmus Sadat Numan

ID: 22-48497-3

Project Proposal

Solix Clothing Brand Full Stack Web Application

Project Title

Solix Clothing Brand Management System

Technologies Used

Backend

- NestJS
- TypeScript
- PostgreSQL
- TypeORM

Frontend

- Next.js
- JavaScript
- Tailwind CSS
- Axios
- React Hooks (useState, useEffect)
- Route Hook for navigation

1. Project Overview

The Solix project is a full stack clothing brand management system developed for handling basic e-commerce operations. The project consists of a secure backend API built with NestJS and a simple frontend application built with Next.js.

The backend system will handle authentication, product management, cart operations, order processing, payment tracking, admin management, and email notifications.

The frontend will connect with the backend APIs and allow users and admins to access and use all backend functionalities through a simple and user-friendly interface. The main focus of the frontend is functionality and API integration rather than advanced UI design.

The goal of this project is to build a practical and scalable full stack web application using modern web technologies.

2. Main Features

Customer Features

- User registration and login
- JWT authentication
- View products and categories
- View product details with images and sizes
- Add products to cart
- Update and remove cart items
- Place orders
- View order history
- Make payment records
- Receive email notifications

Admin Features

- Manage categories
- Manage products
- Upload product images

- Manage product variants and stock
- View users and orders
- Update order status
- Update payment status
- View admin dashboard analytics
- Send notification emails

3. Backend Functional Requirements

Feature	Endpoint Example	Description
Authentication	/auth/register, /auth/login	Register and login using JWT
User Profile	/auth/me	Get authenticated user information
Category Management	/categories	Create and manage product categories
Product Management	/products	Create, update, delete, and view products
Product Images	/products/:id/images	Upload product images
Product Variants	/product-variants	Manage size, stock, and variant price
Cart System	/cart	Add, update, and remove cart items
Order System	/orders	Create and manage orders
Payment System	/payments	Create and update payment records
Admin Dashboard	/admin/dashboard	Revenue and sales analytics
Mail Service	/mail/send	Send email notifications

4. Frontend Functional Requirements

The frontend will be developed using Next.js with JavaScript and Tailwind CSS. The UI design will be simple, clean, and functional.

The frontend will consume all backend APIs using Axios and provide access to all major system features.

Frontend Features

- Login and registration pages
- Product listing page

- Product details page
- Cart management page
- Checkout page
- Order history page
- Admin dashboard page
- Product and category management pages

Frontend Technologies & Concepts

- JavaScript
- Tailwind CSS for styling
- Axios for API requests
- useState for state management
- useEffect for handling lifecycle operations
- Route Hook for page navigation
- ESLint for code quality
- React Compiler support

5. Database Entities & Relationships

The system will use PostgreSQL database with TypeORM relationships.

Main entities:

- User
- Category
- Product
- ProductImage
- ProductVariant
- Size
- Cart
- CartItem
- Order
- OrderItem
- Payment

Relationships implemented:

- One-to-One
- One-to-Many
- Many-to-One

6. Authentication & Security

The system will implement:

- JWT Authentication
- Role-based access control
- Password hashing using bcryptjs
- Protected API routes using Guards

User roles:

- Admin
- Customer

7. Email Notification System

The system will use Nodemailer and MailService for sending emails.

Emails will be sent for:

- Order confirmation
- Order status update
- Payment updates

SMTP credentials will be managed using .env configuration.

8. Technical Requirements

Category	Technology
Backend Framework	NestJS
Frontend Framework	Next.js
Frontend Language	JavaScript
Styling	Tailwind CSS
Database	PostgreSQL
ORM	TypeORM
Authentication	JWT
Validation	class-validator
File Upload	Multer
API Request	Axios
Email Service	Nodemailer
API Documentation	Swagger
Testing Tool	Postman

9. Project Structure

The backend will follow feature-based modular architecture.

Backend Modules

- Auth Module
- Users Module
- Categories Module
- Products Module
- Product Variants Module
- Cart Module
- Orders Module
- Payments Module
- Mail Module
- Admin Module

The frontend will follow component-based structure using Next.js pages and reusable components.

10. Conclusion

The Solix project is a practical full stack e-commerce application that demonstrates modern backend and frontend development concepts. The project includes secure authentication, database management, REST API development, frontend API integration, cart and order workflow, payment handling, and email notification features.

This project will demonstrate knowledge of full stack web development using NestJS, Next.js, PostgreSQL, JavaScript, and TypeScript in a scalable and organized manner.